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Explainability and interpretability of artificial intelligence use in cybersecurity

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Abstract
With the rapid advancement of internet-connected systems and artificial intelligence (AI), AI models have become integral to cybersecurity, particularly in identifying and mitigating cyberattacks. While these AI-based methods are widely utilised, they often generate results that lack interpretability, posing challenges for security experts, developers, educators and others who are involved in understanding the rationale behind specific outcomes. This interpretability gap has driven the emergence of explainable AI (XAI), a growing field focused on enhancing transparency by providing insights and visualisations that make AI decisions understandable to humans. XAI enhances the interpretability of AI-generated outcomes, enabling security experts to better understand and prioritise potential threats. Moreover, it helps to improve the efficiency and accuracy of threat assessments, thereby strengthening overall cybersecurity operations. To explore XAI's role in this cybersecurity domain, a systematic literature review was conducted to offer a comprehensive analysis of its applications in cybersecurity. Adhering to PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines, a structured search was performed on the Scopus database, identifying 565 peer-reviewed studies. After a thorough selection process, 67 relevant papers were analysed. The findings reveal various AI subsets and XAI techniques employed for cybersecurity, challenges associated with XAI applications and future research directions in this domain. This study contributes significantly to the existing body of knowledge as it deepens understanding of XAI's role in cybersecurity. Furthermore, it offers valuable insights for advancing research and practical implementations. This work helps to bridge the gap between AI's technical complexity and its applicability in real-world cybersecurity scenarios.

Keywords Explainable AI, Cybersecurity, Artificial intelligence, Cyberattacks, PRISMA

1 Introduction

Cybersecurity refers to the practice of protecting computer systems, networks, devices and data from unauthorised access, exploitation, modification, or destruction [1]. It encompasses a wide range of technologies, processes and practices designed to safeguard digital assets and ensure the confidentiality, integrity and availability of information [2, 3]. In today's globalisation, cyber civilisation has increased and become an inevitable source of information, transforming society, business and economies in unprecedented ways [4]. Unfortunately, this exponential growth in the use of cyberspace has resulted in an increase in cybercriminal activities [5]. As individuals, businesses, governments and organisations store and exchange vast amounts of sensitive information online, they become potential targets for cyberattacks. Some of the common cyberattacks on and threats to an organisation's systems and networks or that an individual experiences are hacking, malware, phishing and ransomware [4]. In addition, security incidents in the field of cybersecurity include Structured Query Language (SQL) injection attacks, privilege escalation, advanced persistent threats, password attacks, cryptojacking attacks, web application attacks and man-in-the-middle attacks [6–8].

In response to these security incidents, cybersecurity tools based on artificial intelligence (AI) have emerged to assist security teams in effectively mitigating risks and enhancing overall security measures. In addition, cybersecurity researchers have begun exploring AI-based approaches, including machine learning (ML), deep learning (DL), natural language processing (NLP), expert systems, generative AI (Gen AI), computer vision and fuzzy logic [9, 10]. These techniques are being adopted over traditional methods, such as firewalls and signature-based intrusion detection systems (IDSs), to improve the detection accuracy of cyberdefence systems. However, despite the promising performance of AI models in cybersecurity applications, a significant challenge persists: many of these models operate as black boxes [9]. Their internal decision-making processes lack transparency, making it difficult for human operators to understand or trust their outputs [6]. This lack of explainability not only undermines user confidence, but also poses risks in high-stakes environments, where understanding the rationale behind security decisions is critical for accountability, compliance and ethical oversight [11]. The objective of this study was to conduct a comprehensive review of explainable AI (XAI) techniques as applied to cybersecurity systems. Specifically, the study sought to examine how XAI methods can enhance the interpretability and transparency of AI-driven cybersecurity tools, identify various AI subsets used in cybersecurity applications and examine the challenges and limitations associated with implementing XAI in real-world cybersecurity environments.

While there is a growing body of research on AI applications in cybersecurity, fewer studies comprehensively explore the role of explainability in these systems [12]. Existing literature often focuses on the performance metrics of detection models without adequately addressing how these models make decisions or how those decisions can be made interpretable to users [13]. Moreover, the application of XAI techniques such as local interpretable model-agnostic explanations (LIME) and SHapley Additive exPlanations (SHAP) has been explored in isolated case studies, but a unified framework or comparative analysis across different techniques and use cases remains lacking [14, 15].

Therefore, this paper contributes to the emerging field of XAI in cybersecurity as follows: It starts by expanding the current literature by systematically mapping AI subsets and categories of cyberthreats, thereby creating a taxonomy that enhances conceptual clarity. Within this paradigm, the paper offers a systematic examination of several XAI techniques and evaluates their relevance across different cybersecurity contexts. Second, it advances discourse beyond performance metrics by critically examining how interpretability, transparency and accountability can be embedded into AI-driven cyberdefence systems, thereby enhancing trust, reliability, and practical application. Lastly, it discusses the challenges associated with implementing XAI in real-world cybersecurity environments and outlines promising avenues for future research. These contributions not only synthesise and arrange the state of the art but also offer actionable insights and a road map for researchers and professionals working at the intersection of AI, XAI and cybersecurity.

Accordingly, the study was guided by the following four research questions (RQs):

RQ1. What are the AI subsets and different cybersecurity classes?

RQ2. How is XAI used in cybersecurity?

RQ3. What are the challenges associated with XAI applications in cybersecurity?

RQ4. What are the future research directions for applying XAI in the cybersecurity domain?

The remainder of this paper is structured as follows. Section 2 outlines the research methodology employed. Section 3 presents the findings and discusses them in relation to the research questions. Section 4 highlights the contributions of the study. Section 5 addresses the study's limitations and proposes directions for future work, . , and Sect. 6 concludes the paper.

2 Methods

A systematic literature review (SLR) was carried out to examine the application of various XAI techniques in cybersecurity, features and datasets used in the cybersecurity domain. An SLR employs structured methods to collect, analyse and interpret secondary data that is directly related to the research question [16]. This process allows for the collection of pertinent evidence on a given topic that meets predefined eligibility criteria, providing answers to the formulated research questions [17]. This method was therefore chosen to gather and summarise current evidence on various XAI applications used in cybersecurity. The SLR was guided by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines to ensure methodological rigour and transparency. PRISMA is a widely recognised framework for conducting and reporting systematic reviews [17]. It provides a structured approach to identify, screen and select relevant studies while maintaining clarity and reproducibility.

2.1 Search strategy

The data for this study was retrieved from the Scopus database on 8 February 2025. Scopus was chosen for its extensive coverage, high-quality content and advanced analytical tools [18]. Baas, Schotten, Plume, Côté and Karimi [19] emphasise that broader coverage is advantageous for mapping smaller research areas. This is particularly relevant to the field of XAI, as broader coverage facilitates a deeper understanding of its complexities, diverse methodologies and emerging trends. The articles retrieved through the database search were assessed and screened for eligibility using the following basic search string:

("explainable AI" OR "XAI" OR "explainable artificial intelligence" OR "explainability" OR "interpretability") AND ("machine learning" OR "deep learning" OR "computer vision" OR "natural language processing" OR "fuzzy logic" OR "expert systems" OR "generative AI") AND ("cyber security" OR "cyber-security" OR "information security" OR "cyber security" OR "cybersecurity").

2.2 Inclusion and exclusion criteria

The inclusion and exclusion criteria are presented in Table 1. The subject area of the search was limited to the disciplines "Computer Science", "Engineering", "Decision Science" and "Multidisciplinary". In addition to using the search string to identify articles, further criteria were then applied to refine the search results to include records that met the following conditions: (i) authored exclusively in English; (ii) of all types, excluding abstract, correction, data paper, book review, lecture notes, proposal and letter and (iii) publications that employ XAI, ML, DL, NLP, fuzzy logic, computer vision, expert systems and generative AI techniques. The duration period for the literature search was not explicitly defined - this was to ensure a comprehensive and unbiased collection of relevant studies. However, after conducting the search, it became evident that most articles addressing the use of XAI in cybersecurity were published between 2014 and 2025. This period represents a surge in research interest due to advancements in XAI use and increasing applicability in cybersecurity settings. By narrowing the focus to this timeframe, the review captured the most current and relevant developments in the field, ensuring that the findings reflect contemporary practices and innovations.

2.3 Selection process

As illustrated in Fig. 1, the PRISMA framework incorporates a checklist and a flow diagram to facilitate transparent and thorough reporting of research processes. The framework follows three key phases: identification, screening, and included [17, 20, 21]. During the identification phase, the search string detailed in Sect. 2.1 was employed to retrieve articles from the Scopus database. In the screening phase, an Excel worksheet was used to document all extracted papers. Each article's title and abstract were then carefully reviewed to determine its relevance to the application of XAI in cybersecurity. Articles with abstracts that did not align with this theme were excluded from further consideration. Thereafter, full-text articles were sought and thoroughly analysed to confirm their relevance, ensuring that they addressed at least one of the research questions underpinning this systematic review. Initially, 565 articles were retrieved from the Scopus database. Automation tools filtered out 205 articles; 189 articles were excluded based on title and keyword criteria, and 85 more were removed after abstract evaluation. In addition, 9 non-English articles were discarded, and 10 articles misaligned with the thematic focus were excluded. Ultimately, 67 articles were deemed relevant and included in the final review.

Table 1 Inclusion and exclusion criteria

S/No.	Inclusion	Exclusion
1	The period for the literature search is 2014–2025	Publication that is a doctoral symposium
2	The study is written in English	The full text of the publication is not available
3	The focus of the publication is on XAI application in the cybersecurity domain	Research that does not employ XAI, ML, DL, NLP, fuzzy logic, computer vision, expert systems and generative AI techniques
4	The study is peer reviewed.	The paper is less than 6 pages in length
5	The full text of the study is available	The publication is an abstract, correction, data paper, book review, lecture note, proposal or letter
6	Subject areas: Computer Science, Engineering, Decision Science and Multidisciplinary	

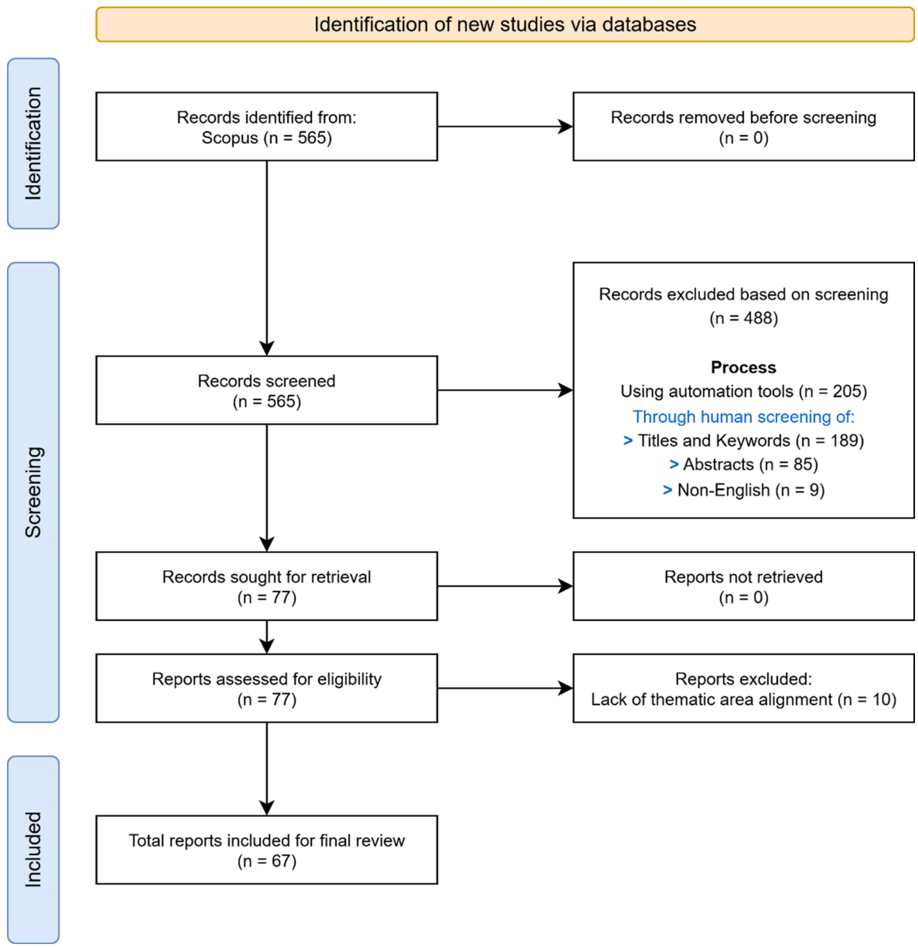

Fig. 1 PRISMA flow diagram illustrating literature search and selection for the application of XAI in cybersecurity [21]

3 Findings and discussion

This section presents a descriptive analysis of the papers extracted through the PRISMA process. An analysis is then provided of the retrieved papers based on the guiding research questions.

3.1 Description of papers based on year of publication

Table 2 presents the annual publication count from 2014 to 2025, summarising the number of research papers published each year, along with the key authors and their respective contributions. Two papers were published in 2014, making up 2.99% of the total research output. The years 2015, 2016 and 2019 had relatively low paper counts, each contributing only 1.49% of the total research output. This might indicate either fewer researchers focusing on the topic, or limited research being published in those years. However, a significant increase in research output was observed in 2017 and 2018, each with 4 papers, accounting for 5.97% of the total research output. The period from 2021 to 2025 marks a notable surge in research activity, particularly in 2021 and 2024, where the number of papers published increased sharply, contributing 13.4% and 19.4%, respectively. This upward trend suggests a growing interest in the topic and advancements in

Table 2 Distribution of reviewed articles included in the study by year of publication

Year of publication	Studies/Authors	Paper count	
		Number of papers	Percentage count
2014	[22, 23]	2	2.99%
2015	[24]	1	1.49%
2016	[25]	1	1.49%
2017	[26–29]	4	5.97%
2018	[10, 30–32]	4	5.97%
2019	[33]	1	1.49%
2020	[34–36]	3	4.48%
2021	[37–45]	9	13.4%
2022	[12, 46–57]	13	19.4%
2023	[58–64]	7	10.45%
2024	[65–77]	13	19.4%
2025	[78–86]	9	13.4%

Table 3 Summary of reviewed research papers by document type

Document type	Number of articles	Percentage
Journals	40	59.7%
Conference proceedings	24	35.8%
Book chapter	2	3%
Book	1	1.5%

the field, with research output accelerating from 2021 onwards and continuing through 2025.

3.2 Description of papers based on document type

The 67 records obtained were subsequently classified into 4 major document types (as depicted in Table 3). Most were journal articles (59.7%), followed by conference proceedings (35.8%), book Chaps. (3%) and books (1.5%).

3.3 AI subsets and cybersecurity classes

The relationship between AI subsets and different cybersecurity classes lies in how various AI techniques enhance security measures across different domains [22]. AI subsets, including ML, DL, NLP, computer vision, expert systems, fuzzy logic and generative AI, provide advanced threat detection, anomaly detection and automated response mechanisms to strengthen these security domains [23]. In response to RQ1, the AI subsets and different cybersecurity classes are discussed in this subsection. These are further summarised in Table 4.

3.3.1 ML

ML, a key subset of AI, has proven effective in building security models using historical cybersecurity data [24]. Techniques such as RF, SVM, KNN, Naïve Bayes and DT have been widely applied. In particular, studies by Choubisa, Doshi, Khatri and Hiran [25] and Reddy, Kumar, Rao and RajaRajeswari [26] demonstrate that RF algorithms significantly enhance intrusion and malware detection systems, illustrating RF's versatility across different threat domains. These findings align with the work of Ghanem, Aparicio-Navarro, Kyriakopoulos, Lambotharan and Chambers [27], who propose using SVMs alongside an unsupervised anomaly-based IDS to reduce false alarms and enhance detection

Table 4 AI subsets and different cybersecurity classes

S/No.	AI subset	AI technique	Cybersecurity class	Studies/Authors
1	ML	Random forest (RF)	Intrusion detection	[46]
2		RF	Malware detection	[58]
3		Support vector machine (SVM)	Anomaly detection	[26]
4		SVM	Malware detection	[22]
5		K-nearest neighbour (KNN)	Network intrusion detection	[65]
6		KNN	Anomaly detection	[66]
7		Naïve Bayes	Intrusion detection	[37]
8		Decision tree (DT)	Malicious behaviour analysis	[67]
9		Naïve Bayes, RF, SVM and DT	Malware detection	[68]
10	DL	Convolutional neural networks (CNNs)	Network intrusion detection	[47]
11		Multi-layer perceptron (MLP)	Anomaly detection	[27]
12		Deep belief networks (DBNs) and stacked autoencoders (SAEs)	Anomaly detection	[30]
13		Recurrent neural networks (RNNs)	Anomaly detection	[48]
14		Long short-term memory (LSTM)	Anomaly detection	[33]
15		CNN- LSTM	Malware detection	[49]
16	NLP	Sentiment annotations Wordnet	Phishing detection	[23]
17		Bag-of-Words (BoW)	Fraud detection	[38]
18	Computer vision	Generative adversarial networks (GANs)	Network traffic anomaly detection	[59]
19		Speed up robust features (SURF)	Phishing detection	[24]
20		Histograms of oriented gradients (HOG)	Phishing detection	[25]
21	Fuzzy logic	Fuzzy logic inference system (FLIS)	Risk detection	[31]
22		Fuzzy logic-based cybersecurity enhancement	Enhancing cybersecurity strategies	[77]
23	DL and expert systems	Deep embedded neural network expert system (DeNNeS)	Network traffic anomaly detection	[50]
24	Generative AI	Generative adversarial networks (GANs)	Network intrusion detection system	[51, 69]

accuracy. This complements RF's classification strengths with improved detection accuracy. However, Biggio, Corona, Nelson, Rubinstein, Maiorca, Fumera, Giacinto and Roli [28] examined SVM applications in malware detection and highlighted vulnerabilities such as poisoning, evasion and privacy breaches, demonstrating both the strengths and security risks of SVMs in cybersecurity contexts. This highlights a critical limitation not observed to the same extent in RF-based studies.

Similarly, studies have demonstrated the effectiveness of KNN in cybersecurity applications. Agarwal, Gill, Chauhan, Kapruwan and Banerjee [29] found KNN to be robust in classifying network security attacks, while Zhao, Zhang, Zhou and Lou [30] enhanced KNN-based anomaly detection through optimised attribute and distance weighting, directly addressing the accuracy limitations faced in earlier models. This progression in KNN research echoes developments seen in Naïve Bayes applications. Fors instance, Majeed, Abdullah and Mushtaq [31] successfully applied Naïve Bayes in IoT-based drone security model, successfully identifying cyberattacks by analysing sensor and network data patterns.

DT models also contribute effectively in cybersecurity, with Jamil, Creutzburg and Nafiz Aydin [32] highlighting their value in protecting critical infrastructure through supervised learning. From a comparative perspective, Kothari and Tidke [33] compared DT alongside multiple ML techniques including Naïve Bayes, RF, and SVM for malicious URL detection, highlighting their advantages over traditional blacklist methods and emphasising the need for more adaptive detection systems. Collectively, these studies show that ML techniques are not siloed to specific tasks but often overlap and complement each other, with RF and SVM excelling in structured detection, KNN and Naïve Bayes adapting to anomaly-based and IoT contexts, and DT providing interpretability and robustness in infrastructure protection.

3.3.2 DL

DL has been increasingly adopted in cybersecurity because of its ability to extract complex patterns from large-scale and high-dimensional data. Different DL architectures have been tailored to address unique cybersecurity challenges, with each offering distinct advantages. For instance, Nedeljkovic and Jakovljevic [34] applied CNNs to develop cyberattack detection algorithms for industrial control systems (ICS). Their CNN-based model analyses network traffic to identify malicious patterns and demonstrates high detection accuracy with minimal false positives. Trained on real-world ICS datasets, the model shows strong practical relevance for enhancing ICS cybersecurity. This highlights the strength of CNN-based architectures in spatial pattern recognition within structured network traffic. Similarly, Akhtar and Feng [35] extended CNN applications by combining them with LSTM in a hybrid CNN-LSTM DL model for real-time malware detection. In this setup, the CNN extracts spatial features from malware binaries or network traffic, while the LSTM captures sequential patterns. This combined approach improves detection accuracy and efficiency, outperforming traditional machine learning and standalone deep learning models with reduced false positives.

Other DL approaches focus on sequential and flow-based data. Van Efferen and Ali-Eldin [36] explored an MLP neural network for anomaly detection in network traffic using flow-based analysis. Unlike traditional packet-based approaches, their MLP model analyses flow-level features to distinguish between normal and malicious activity. Evaluated on benchmark datasets, the model demonstrated high accuracy, low false positives and strong scalability, making it suitable for real-time cybersecurity applications. In parallel, RNN-based models for network and cybersecurity applications explored by Yenduri and Gadekallu [37], which include LSTM and gated recurrent unit (GRU) architectures, showcase the importance of sequential time-series modelling for evolving and persistent threats. These models effectively detect anomalies and cyberthreats such as DDoS attacks, malware intrusions and phishing with greater accuracy than traditional rule-based systems, supporting real-time threat mitigation.

Gasmi, Laval and Bouras [38] further advanced sequential modelling through LSTM networks for extracting cybersecurity information from unstructured textual data. Their model effectively classifies and identifies domain-specific terms with minimal false positives, outperforming traditional NLP methods. This approach highlights applications in automated cybersecurity reporting, knowledge graph creation and real-time threat analysis, enhancing the organisation and analysis of threat intelligence.

DL research has also addressed the challenge of adapting to dynamic environments such as 5G networks. Maimó, Gómez, Clemente, Pérez and Pérez [39] developed a self-adaptive anomaly detection system for 5G mobile networks that dynamically adjusts computational resources to manage unpredictable traffic fluctuations. Their approach uses a two-level deep learning model combining DBN and SAE deployed within the radio access network (RAN). Tested on Bonet and CTU datasets, the system enhances real-time detection accuracy and adapts to new attack behaviours. This adaptability complements the strengths of CNN and RNN architectures by ensuring scalability and responsiveness in rapidly changing contexts.

Taken together, these studies highlight how DL techniques are not isolated solutions but complementary approaches. CNNs excel at identifying static spatial features, RNNs and LSTMs address temporal and sequential patterns, hybrid architectures (such as CNN-LSTM models) bridge spatial-temporal gaps, and self-adaptive models support scalability in next-generation networks.

3.3.3 NLP

NLP has proven valuable in detecting cyberthreats embedded within textual communication, particularly phishing and fraudulent emails. Aggarwal, Kumar and Sudarsan [40] focused on detecting phishing emails that do not contain links or URLs, specifically analysing email communication. Their use of WordNet and Stanford CoreNLP demonstrates how semantic and syntactic analysis can reveal subtle social engineering cues often overlooked by traditional filters. Similarly, Hamisu and Mansour [41] developed a fraudulent email classifier to detect and classify Advance Fee Fraud (AFF) emails using an NLP approach known as the Bag-of-Words (BoW) model. The study demonstrated that the proposed classifier, utilising the BoW model, effectively detected and classified AFF emails, thereby contributing to enhanced cybersecurity measures against such fraud. Together, these studies illustrate the adaptability of NLP approaches across varying levels of linguistic complexity, from nuanced semantic analysis to efficient lexical-based classification, thereby broadening their applicability in combating phishing and fraud.

3.3.4 Computer vision

Computer vision techniques extend cybersecurity beyond textual or numerical data by leveraging visual representations to detect threats. Ranka, Shah, Vora, Kulkarni and Patil [42] introduced a novel cybersecurity approach using computer vision to transform network traffic into visual data for distinguishing legitimate from malicious activities. Their method integrates GANs to generate synthetic data that addresses dataset scarcity while enabling DL models to recognise malicious patterns in real time. Similarly, Rao and Ali [43] used computer vision with SURF feature detection to compare website screenshots for improved phishing detection, including zero-day attacks. Bozkir and Sezer [44] further advanced this area by using HOG to analyse webpage visuals and detect phishing by capturing page layout signatures, aiding in identifying novel attacks. These studies demonstrate the potential of computer vision in cybersecurity. Collectively, they highlight how computer vision bridges gaps in conventional approaches by addressing both data scarcity and novel attack vectors.

3.3.5 Fuzzy logic

Fuzzy logic offers a valuable approach for handling uncertainty and imprecision in cybersecurity contexts where traditional binary or statistical methods may fall short.

Alali, Almogren, Hassan, Rassan and Bhuiyan [45] investigated the use of FLIS to improve cybersecurity risk assessment, showing how ambiguity in threat data can be more effectively managed through flexible decision-making rules. Imamguluyev, Huseynli and Hajiyev [46] built on this approach by exploring how fuzzy logic, inspired by human reasoning, enhances cybersecurity strategies by enabling adaptive, self-learning and proactive defences that dynamically respond to evolving threats, vulnerabilities and security effectiveness. Together, these studies illustrate how fuzzy logic not only supports nuanced assessment of incomplete information, but also enables systems to continuously learn and adjust, thereby offering resilience in highly variable cybersecurity contexts.

3.3.6 DL and expert systems

The integration of DL and expert systems has emerged as a promising avenue for achieving both automation and interpretability in cybersecurity. Mahdavifar and Ghorbani [47] present an advanced cybersecurity framework that leverages DL and expert systems to enhance cyberattack detection. The DeNNeS model integrates a DNN with an expert system, allowing for precise and automated detection of cyberthreats in real time. By embedding domain-specific knowledge into the system, the model improves interpretability and decision-making in threat detection. This hybrid approach bridges the gap between high-performing but opaque DL models and interpretable knowledge-driven systems, offering a pathway toward more reliable and trustworthy cybersecurity solutions.

3.3.7 Generative AI

Generative AI has gained traction in cybersecurity for its ability to address data scarcity and imbalance, which often hinder the training of robust IDSs. Park, Lee, Kim, Park, Kim and Hong [48] developed an AI-driven NIDS that addresses data imbalance by using GANs to create synthetic network traffic, enhancing classification accuracy. Their autoencoder-based models with DNNs and CNNs outperformed traditional methods. In a similar approach, Zhao, Fok and Thing [49] used multiple GAN variants to generate realistic anomalous traffic patterns, improving NIDS performance, especially for attacks with limited training data. Collectively, these studies highlight GANs' role in augmenting datasets, improving detection of rare and emerging threats, and increasing the resilience of IDS against evolving cyberattacks.

3.4 The use of XAI in cybersecurity

XAI is increasingly being adopted in cybersecurity to improve transparency, trust and interpretability of AI-driven security systems [14]. XAI helps stakeholders understand why AI models flag activities as threats or anomalies [50], which is vital in areas like threat detection, malware analysis and intrusion detection [11]. XAI techniques are categorised as post hoc (explanations after prediction), and can provide local (individual prediction) or global (overall model behaviour) insights [51]. Many methods are model-agnostic, applicable across different AI models treating them as black boxes [52]. Common techniques include SHAP, LIME, counterfactual explanations and anchors [53–55],. In addition, visualisation methods (such as heatmaps, saliency maps, or partial dependence plots) provide intuitive ways to observe which features or input regions influence predictions [56]. Another notable approach is LEMNA, designed for high-dimensional and security-related datasets, which uses mixture regression models for more accurate explanations. Collectively, these methods help in demystifying AI decision-making by highlighting the most influential features and clarifying the reasoning process behind predictions, thereby improving transparency, accountability, and user trust in AI systems [57]. In response to RQ2, the use of XAI in cybersecurity is discussed in this subsection. Several XAI techniques are now being utilised in cybersecurity to enhance the interpretability and transparency of machine learning models. These are summarised in Table 5.

3.4.1 SHAP

SHAP-based methods provide global and feature-level explanations by quantifying each feature's positive or negative contribution to model outputs using Shapley values [50]. Alenezi and Ludwig [52] developed an explainable ML framework for malware and malicious URL detection, integrating SHAP techniques (KernelExplainer, TreeExplainer, DeepExplainer) to interpret models like RF and XGBoost, tested on ISCX-URL2016 and Android malware datasets. Giudici and Raffinetti [51] utilised an enhanced version of SHAP called the Shapley–Lorenz decomposition method to improve global interpretability and robustness, applying it to ordinal cyber-risk classification.

3.4.2 LIME

LIME provides local interpretability by approximating complex models around individual instances, helping explain specific predictions [50]. Alodibat, Ahmad and Azzeh [58] applied LIME with a DT classifier on the BIG 2015 malware dataset, demonstrating effective, explainable malware detection. Similarly, Baniya, Vinod, Chinta, Vishnu and Thangam [59] used LIME to identify key features influencing predictions in intrusion detection systems, with RF and adaptive boosting models, underscoring their effectiveness in cybersecurity.

3.4.3 SHAP, LIME and rulefit

Abou El Houda, Brik and Khoukhi [60] developed a deep learning-based intrusion detection system for IoT networks enhanced with XAI technique: LIME, SHAP for

Table 5 Application of XAI in cybersecurity

XAI techniques	Application in cybersecurity	Post hoc	Global	Local	Model- agnostic	Studies/ Authors
SHAP	Malware detection	x	x	x	x	[39, 52]
LIME	Intrusion detection	x		x	x	[60, 70]
LIME, SHAP and RuleFit	Intrusion detection	x	x	x	x	[53, 54]
Counterfactual explanations	Compliance	x		x	x	[28, 32]
Visualisation techniques (Saliency map, Grad-CAM)	Intrusion detection	x	x	x	x	[29, 34, 40, 41, 55, 61]
LEMNA	Malware detection	x		x	x	[10, 42]
Anchors	Phishing detection	x		x	x	[45]
Anchors, SHAP, counterfactual explanation and LIME	Botnet detection	x		x	x	[57]

local explanations, and RuleFit for global feature importance, improving interpretability of model decisions. Similarly, Sivamohan and Sri [61] proposed a BiLSTM-based XAI intrusion detection system optimised using krill herd optimisation to select relevant features from NSL-KDD [62] and honeypot datasets [63]. They also used LIME and SHAP to provide transparent explanations, boosting trust in the system's outputs.

3.4.4 Counterfactual explanations

Counterfactuals are user-friendly, post hoc local explanation methods that overcome several limitations of previous techniques. They avoid relying on predefined baselines, eliminate complex approximations rooted in game theory and do not require universal feature assumptions ("Generating Plausible Counterfactual Explanations for Deep Transformers in Financial Text Classification"). Wachter, Mittelstadt and Russell [64] introduced the concept of counterfactual explanations within the context of general data protection regulation (GDPR) compliance, empowering users to challenge or understand automated decisions by posing "Why?" or "Why not?" questions. Binns, Van Kleek, Veale, Lyngs, Zhao and Shadbolt [65] also indicate that users tend to prefer counterfactual explanations over feature-importance methods, finding them more relatable and actionable for interpreting model outcomes.

3.4.5 Visualisation techniques

Visualisation tools play a key role in explaining AI decisions in cybersecurity. Naeem, Alshammari and Ullah [66] developed an explainable malware detection framework for IoT using image-based CNN models (Inception-v3) and Grad-CAM heatmaps to visually highlight features influencing malware classification. Techniques such as t-SNE validate feature clustering. Becker, Drichel, Müller and Ertl [67] propose a visual analytics framework to interpret deep learning for domain generation algorithm detection through clustering and DTs. Norton and Qi [68] introduced a web tool for exploring adversarial example generation interactively. Iadarola, Martinelli, Mercaldo and Santone [69] used Grad-CAM to enhance transparency in image-based malware classification, aiding analysts in validating model decisions. Hong, Papazachos, del Rincon and Miller [70] improved intrusion detection by combining feature saliency with Gaussian mixture clustering. Lin, Lee and Celik [71] tested saliency explanation accuracy by embedding backdoor triggers in images, assessing if saliency maps correctly identify misclassification causes. These studies demonstrate the value of visual and interactive methods in making complex ML models interpretable and trustworthy.

3.4.6 LEMNA

LEMNA is an XAI technique tailored for AI-based security, combining mixture regression with fused lasso to produce high-fidelity explanations. Guo, Mu, Xu, Su, Wang and Xing [10] developed LEMNA specifically for malware detection, demonstrating its effectiveness on a PDF malware dataset. Chakraborty, Krishna, Ding and Ray [72] applied LEMNA to interpret vulnerability classifications in code snippets by highlighting meaningful tokens that explain the model's decisions.

3.4.7 Anchors

Kluge and Eckhardt [73] propose a framework that informs users about the specific words and phrases in an email that influenced a phishing detector's classification of the email as suspicious. This is achieved through local perturbations, inspired by the Anchors method. The study identifies key textual elements indicative of phishing attempts, enhancing the interpretability of phishing detection systems. The research further demonstrates that the proposed XAI approach effectively extracts critical linguistic features that differentiate legitimate emails from phishing attempts. These findings suggest that XAI can play a crucial role in developing more effective phishing prevention strategies.

3.4.8 Anchors, SHAP, counterfactual explanation and LIME

Suryotrisongko, Musashi, Tsuneda and Sugitani [74] investigated botnet detection using domain generation algorithms (DGAs) and applied four XAI techniques, namely Anchors, SHAP, counterfactual explanation and LIME, to improve transparency and address trust issues in AI/ML-based cyberthreat intelligence systems. They propose integrating XAI with open-source intelligence (OSINT) to provide clearer insights into AI decisions and enhance threat intelligence.

However, this integration raises challenges such as data quality, legacy system compatibility and increased adversarial risks, discussed in detail in Sect. 3.5.

3.5 Challenges associated with XAI applications in cybersecurity

While XAI has the potential to enhance cybersecurity by making AI systems more transparent and trustworthy, it faces challenges ranging from complexity of explainability in DL to adversarial attack and human biases [23]. In response to RQ3, this section covers the major challenges associated with XAI applications in cybersecurity. These are summarised in Table 6.

3.5.1 Technical-related issues

One of the major challenges associated with XAI applications in cybersecurity lies in handling the complexity of DL models used in explainable IDS [75]. Although these

Table 6 Challenges associated with XAI applications in cybersecurity

Studies/Authors	Technical-related issues	Data-related issues	Ethical issues
[71]	x		
[35]	x		
[43]	x	x	
[72]		x	
[12]	x	x	x
[73]			x
[62]			x
[56]			x
[74]	x		
[36]	x		
[78]	x		
[79]	x		
[80]	x		
[63]	x		
[75]	x		

models deliver outstanding performance, their lack of transparency makes interpretation difficult [76]. Extracting meaningful explanations from such complex models necessitates advanced techniques, often involving a trade-off between accuracy and interpretability [75]. According to Arrieta, Díaz-Rodríguez, Del Ser, Bennetot, Tabik, Barbado, García, Gil-López, Molina and Benjamins [76], XAI can be effective when it provides explanations that are easily understood by cybersecurity professionals and non-experts. However, achieving a balance where explanations are detailed enough to be valuable but simple enough to be actionable without oversimplification is a challenge [77, 78]. In other words, increasing the interpretability of a model can lead to a decrease in its accuracy. Kalutharage, Liu and Chrysoulas [79] propose that the integration of XAI techniques with expert domain knowledge can bridge the gap effectively between intricate AI predictions and human-comprehensible decision-making. This approach not only improves detection accuracy, but also enhances the interpretability of results, making AI-driven insights more transparent and actionable [80–82].

Moreover, many cybersecurity infrastructures still rely heavily on legacy systems (i.e. outdated hardware, software, or network protocols) that were designed long before the advent of modern XAI technologies [83]. These systems often form the backbone of critical industries such as government, healthcare, finance and energy, where continuity, stability and compliance with older regulatory standards have historically taken precedence over rapid innovation. However, these legacy environments present significant integration challenges when it comes to implementing XAI frameworks [80]. For instance, the legacy systems typically lack the computational resources and data pipelines necessary to support modern XAI models, which often require real-time processing, high memory capacity and scalable data handling [81].

In addition, the rapid evolution of cybersecurity threats poses a significant challenge for XAI techniques [12]. Cyberattackers constantly develop new strategies and tactics to bypass traditional and AI-based defence mechanisms [84]. As a result, XAI techniques face this challenge of adapting to these new threats while maintaining transparency and interpretability [12]. For instance, traditional AI models that detect known attack patterns may become less effective as adversaries modify their tactics or introduce new forms of obfuscation [83]. This requires XAI systems to not only explain the reasoning behind detection, but also adaptively integrate new knowledge and retrain to recognise previously unseen attack signatures. Moreover, the integration of adaptive learning capabilities within XAI can be complex [12]. Real-time model updating and retraining to accommodate new threat data can compromise the stability of explanations, making it difficult for users to understand how changes impact decision-making.

3.5.2 Data-related issues

Adversarial attacks involve the intentional manipulation of input data to deceive AI models into making incorrect predictions [83]. Research has demonstrated that even XAI models are susceptible to such attacks [83, 85]. For instance, an image classifier that correctly identifies a stop sign can be misled into labelling it as a speed limit sign by adding subtle noise. In the cybersecurity domain, adversarial inputs can cause models to misclassify malicious traffic as benign, or vice versa [85]. Moreover, in malware detection systems, attackers may not only deceive the model but also manipulate the explanation output, producing misleading feature importance that falsely justifies the incorrect prediction [85]. Attackers can further craft inputs that bypass detection systems while simultaneously distorting XAI-generated explanations, thereby undermining trust and making it more difficult to derive reliable, actionable insights [83].

This vulnerability is further exacerbated when outdated or static datasets are used to train or evaluate XAI systems. According to Zhang, Al Hamadi, Damiani, Yeun and Taher [12], a significant issue with widely used cybersecurity datasets is their lack of updates and completeness. Many public datasets do not incorporate recent types of cyberattacks, which can lead to gaps in training data for AI models, including XAI-based applications. This is often due to privacy and ethical concerns that restrict sharing sensitive and up-to-date data. Consequently, datasets may not reflect emerging threats or newly developed attack vectors, undermining the robustness and efficacy of AI systems in developing effective defensive mechanisms against current cyberthreats. This is particularly problematic for XAI, which aims to provide interpretable insights while adapting to evolving security landscapes [12]. If datasets are not updated to include recent, sophisticated attacks, the XAI systems may be less reliable in offering transparent and accurate explanations that keep pace with modern threat scenarios.

3.5.3 Ethical and privacy concerns

Ensuring that XAI explanations meet regulatory and ethical standards is a significant concern, especially in sensitive domains like cybersecurity [86]. Regulations such as the GDPR mandate transparency and accountability in automated decision-making, requiring that individuals have access to understandable explanations of algorithmic outcomes [87]. However, applying XAI in cybersecurity introduces complex privacy challenges, as the explanations often involve analysis of sensitive network traffic, system logs, or user behaviour patterns [87].

For example, techniques such as SHAP or LIME may highlight specific features, such as IP addresses, access times, or device IDs, that could unintentionally expose confidential system details or user identities when shared with external stakeholders or third-party vendors [88]. In another scenario, revealing which features influenced a model's decision to block a cyberattack could inadvertently disclose security vulnerabilities, offering adversaries insights into how to bypass detection [12].

3.6 Comparative synthesis: traditional AI vs. XAI in cybersecurity

The previous sections have given a comprehensive account of the utility of XAI in cybersecurity, as well as the associated challenges, based on the analysed studies. This section synthesises how XAI compares with traditional AI approaches. While earlier sections highlighted the strengths and limitations of XAI in isolation, it is equally important to position these insights against the established paradigm of traditional AI. Several researchers (e.g. Aggarwal, Sachan, Verma and Dhanda [89], Dwivedi, Dave, Naik, Singhal, Omer, Patel, Qian, Wen, Shah and Morgan [90], as well as Das and Rad [91]) have examined this comparison, making a strong case for how XAI enhances interpretability, trust, and compliance without entirely sacrificing performance. Table 8 provides a consolidated view of this comparison across key dimensions discussed in this review, including interpretability, trust, decision transparency, adaptability, compliance, and performance.

Table 7 Comparative analysis of traditional AI and XAI in cybersecurity across key dimensions (Synthesised from: [37, 49], 89– [91])

Dimension	Traditional Artificial Intelligence	Explainable AI
Interpretability	Decisions derived through traditional AI are often opaque and difficult to interpret.	XAI provides human-understandable explanations for decisions.
Trust	Trust is limited; users may not trust outputs due to lack of explanation or overly technical details.	With XAI, trust is enhanced as users gain confidence through interpretable outputs.
Decision Transparency	Decision-making is opaque, with little insight into the reasoning process.	XAI ensures transparency by providing rationale and feature importance for decisions.
Adaptability	Models can easily adapt to new data, but this adaptability does not extend to explainability.	XAI is moderately adaptable, with some constraints introduced by interpretability requirements.
Compliance	Meeting compliance requirements is challenging without clear decision rationale.	XAI improves compliance by offering transparent decision logic that supports regulatory requirements.
Performance	Optimised for accuracy and speed, often at the cost of interpretability.	Slight trade-off in speed/accuracy for interpretability, though hybrid models are reducing this gap.

Table 8 Future research directions for applying XAI in the cybersecurity domain

S/No.	Future Research Directions	Studies/Authors
1	High-quality datasets	[12, 81–83]
2	Responsible AI (RAI)	[74, 84, 85]
3	User-centred XAI	[12]
4	Handling uncertainty in XAI for cybersecurity	[64, 76, 86]
5	Adversarial attacks and defences	[43, 64, 72]
6	Universal framework and formalisation	[44, 74]

As shown in Table 7, while traditional AI excels in raw performance and adaptability, its most critical limitation in cybersecurity is its black box nature, which restricts interpretability and undermines trust [9, 52]. XAI addresses these limitations by introducing transparency and interpretability, which are particularly critical in high-stakes environments such as cybersecurity. Furthermore, XAI offers trust and compliance benefits critical for cybersecurity contexts, thus further strengthening its relevance. This trade-off highlights the need for hybrid approaches that balance accuracy with transparency and explainability.

Therefore, in cybersecurity, where explainability is as vital as accuracy, XAI represents a paradigm shift from opaque prediction models toward accountable and trustworthy decision-making.

3.7 Future research directions for applying XAI in the cybersecurity domain

The future of applying XAI in the cybersecurity domain encompasses several promising research directions to enhance its effectiveness, interpretability and real-world applicability. In response to RQ4, this section deals with the future research direction associated with XAI applications in cybersecurity. These are summarised in Table 8.

3.7.1 High-quality datasets

According to Zhang et al. [12], future research should focus on creating high-quality, comprehensive and current datasets for XAI applications in cybersecurity. This is because the effectiveness of XAI methods in cybersecurity systems relies heavily on the quality and comprehensiveness of the datasets used [92]. However, data from networks or the broader internet usually includes sensitive information such as personal or corporate details, which, if released publicly, might expose vulnerabilities in the originating systems [93]. In addition, dataset imbalances in both volume and features can negatively impact the development of XAI-based cybersecurity solutions [94].

3.7.2 Responsible AI (RAI)

Hassija, Chamola, Mahapatra, Singal, Goel, Huang, Scardapane, Spinelli, Mahmud and Hussain [78] suggest RAI as one of the future research directions for the cybersecurity domain. RAI extends beyond the scope of XAI by embedding broader societal and ethical considerations into the development and deployment of AI systems [95]. Unlike XAI, which focuses primarily on making AI decisions understandable to users, RAI ensures that AI systems align with the ethical, legal and moral standards of society [96].

3.7.3 User-centred XAI

Recent research has highlighted the importance of user-centred XAI systems, particularly focusing on making explanations more comprehensible to users across various domains, including cybersecurity [12]. A user-centred approach prioritises explanations that align with human cognitive processes and preferences, aiming for high user satisfaction, better understanding and improved decision-making outcomes [97].

3.7.4 Handling uncertainty in XAI for cybersecurity

Due to the inherently uncertain and constantly changing nature of the cybersecurity field, it is essential to develop XAI methods that can manage uncertainty effectively [97]. One such approach involves using probabilistic models, such as Bayesian networks, which allow for representation and reasoning under uncertainty [98]. In addition, robust optimisation techniques offer a structured way to handle uncertainty in decision-making processes [99]. Overall, creating XAI methods capable of managing uncertainty is a significant challenge that needs to be met to facilitate broader acceptance and integration of XAI in cybersecurity applications [97].

3.7.5 Adversarial attacks and defences

While recent studies have explored cyberthreats and defensive strategies targeting AI model performance, adversarial attacks and defences specifically addressing the explainability of XAI models remain an area requiring further investigation [85, 97]. In addition, Kuppa and Le-Khac [83] emphasise that more attention should be given to adversarial inputs in sample data because they pose significant challenges to the robustness and reliability of XAI.

3.7.6 Universal framework and formalisation

Hassija, Chamola, Mahapatra, Singal, Goel, Huang, Scardapane, Spinelli, Mahmud and Hussain [78] suggest that future research should focus on developing a unified framework to close the gap between interdisciplinary studies on how people explain and formalise patterns in algorithmic forms and the rapidly evolving XAI techniques and frameworks. This proposed framework would formalise XAI by leveraging its two foundational pillars, namely explanation and interpretation. Moreover, it could serve as a unifying agent to manage the growing heterogeneity in the field and consolidate the diverse methods being developed [100].

4 Contributions of the study

This study makes several distinct contributions to the field of explainable artificial intelligence (XAI) in cybersecurity. These contributions, as listed below, provide both a theoretical foundation and practical roadmap for advancing XAI in cybersecurity, supporting both scholarly research and applied system development.

i. Taxonomy of XAI techniques: The study presents a structured taxonomy of widely used XAI techniques, including SHAP, LIME, counterfactual explanations and others, and critically analyses their suitability for various cybersecurity tasks. This categorisation enables a clearer understanding of how each technique aligns with specific interpretability needs in threat detection, malware analysis and other security functions.
ii. Identification of key challenges: The study highlights major challenges associated with implementing XAI in cybersecurity environments, such as the technical, data-related issues, ethical concerns and the difficulties in achieving real-time explainability. These challenges are discussed in the context of operational constraints, offering insights into practical limitations and research gaps.
iii. Practical guidance through real-world applications: The study provides illustrative examples of real-world XAI applications in cybersecurity, such as intrusion detection systems and phishing detection tools. These use cases serve to guide practitioners and system designers in selecting and adapting XAI techniques suited to specific operational challenges.
iv. Emphasis on data representativeness: The study emphasises the critical role of representative and context-specific data in ensuring the effectiveness and trustworthiness of XAI-driven cybersecurity systems. It highlights that without appropriate data, even the most sophisticated XAI techniques may fail to deliver reliable insights.

5 Limitations of the study and future research

Validating the research methodology is fundamental to ensuring the credibility of findings. One of the main threats to the validity of this study is the potential omission of relevant papers during the selection process, as well as biases that may have arisen during data extraction. The Scopus scholarly database was used exclusively, which, while comprehensive, does not encompass all potentially relevant studies. Other databases might contain significant work that could influence the outcomes. As a result, the findings of this study cannot be entirely generalised across the broader research landscape. However, the study offers significant insights into the application of XAI in cybersecurity, emphasising its relevance within the field. This is particularly relevant since the initial search indicated that Scopus contained the majority of the necessary publications, making it a reliable starting point for the research.

Future research should expand the scope of database coverage to include additional sources beyond the Scopus database, capturing a broader spectrum of relevant literature. This approach would enrich the diversity and comprehensiveness of the findings. In addition, while the literature search period was not explicitly defined to avoid bias, analysis revealed that most articles on XAI in cybersecurity were published between 2014 and 2025. This limitation may hinder the study's relevance in a rapidly evolving field. Consequently, future research should periodically revisit and update the review to reflect advancements in XAI beyond this timeframe, ensuring a more dynamic and current understanding of the domain. Furthermore, future studies should refine and expand search strings to incorporate additional terms, synonyms and variations relevant to XAI and cybersecurity. This enhancement would ensure a more thorough exploration of the literature, uncovering insights that might otherwise be overlooked.

Furthermore, it is recommended that the current lack of technical comparisons, such as classifier performance and dataset robustness, be addressed. These are largely missing from the existing narrative.

Moreover, future research could also consider a comparative matrix that would map various XAI techniques to specific cybersecurity applications, along with their corresponding explanation levels (local or global), evaluation metrics and datasets used. This would provide a structured overview to support systematic comparisons and identify areas where empirical gaps remain.

6 Conclusion

This study highlights the critical importance of XAI in mitigating the interpretability limitations inherent in AI-driven cybersecurity systems. Through a systematic literature review guided by PRISMA methodology, the research offers a comprehensive analysis of XAI applications examining the range of techniques employed, the implementation challenges encountered and the directions proposed for future direction. Key findings reveal that XAI not only enhances the transparency and trustworthiness of AI models in cybersecurity, but also serves as a practical bridge between complex algorithmic decisions and human-centred understanding. This dual contribution (advancing both theoretical insight and practical applicability) positions XAI as a cornerstone in the development of responsible and effective cybersecurity solutions.

Moreover, the study contributes a synthesised taxonomy of techniques, identifies various XAI techniques and categorises challenges into technical, data-related and ethical/legal domains. These insights are valuable for researchers, developers and cybersecurity practitioners seeking to design interpretable, transparent and resilient AI systems. Looking ahead, the study identifies several promising avenues for future work, including the refinement of explanation methods for deep learning models, improved evaluation metrics for XAI in security contexts and the development of more diverse, representative datasets. Addressing these gaps will be crucial to ensuring that XAI continues to evolve in step with the growing sophistication of cyberthreats.

Ultimately, this review lays a strong foundation for ongoing research and practical innovation, supporting the advancement of AI systems that are not only intelligent but also explainable, ethical and secure.

Author contributions
All authors contributed to the conceptualisation of the study. L.O. was responsible for the original draft preparation, methodology development, and preparation of figures and tables, as well as the analysis and interpretation of results. T.B. and S.M. provided supervision and contributed to the review and editing of the manuscript.

Funding
No specific funding was received to carry out this study.

Data availability
No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

This study made use of secondary data derived from research articles retrieved from the Scopus academic database. Ethical clearance exemption for the use of secondary data was granted by the Research Ethics Committee of the School of Consumer Intelligence and Information Systems at the University of Johannesburg (Clearance Number: SCiiS2025101).

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 1 May 2025 / Accepted: 15 October 2025
Published online: 28 October 2025

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